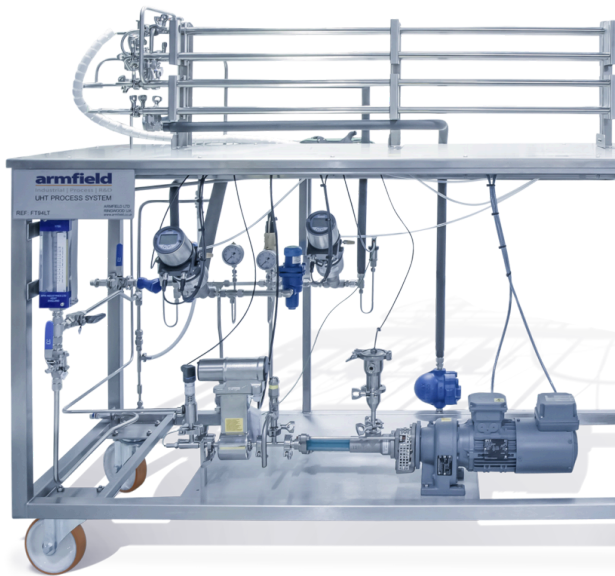




FT94LT HTST/UHT Mini Pilot System

Due to the increased demands from marketing departments, food and beverage manufacturers' R&D teams are required to prepare increasingly larger volumes of sample product. As a result, Armfield have developed the new FT94LT Mini Pilot System complete with Tubular Heat Exchanger. The new FT94LT is a cost-effective HTST/UHT processing system, based on a tubular heat exchanger only, with a range of options including data logging, to tailor the product to your requirements.

Suitable for various industries including media sterilisation, baby foods, beverages, dairy, and pharmaceuticals.

PRODUCT CODE: FT94LT

Categories: Armfield Beverages Equipment, Armfield Dairy, Armfield Educational Food Technology, Armfield Liquid Foods, Armfield Nutraceuticals, Armfield Pharmaceuticals, FT Series - Educational Food Technology, FT94LT HTST/UHT Mini Pilot System, IFT Industrial Food Technology, IFT Series - Industrial Food Technology

Tags: 316 Stainless Steel , CIP , Continuous Operation , Derivative , Electro-Pneumatic Steam-Control Valve , Feed Pump , FT140 , FT52 , FT83 , FT94LT , Heat Exchanger , Homogeniser , HTST , Mini Pilot System , PID , Pilot System , PLC Control , Preheat , Steam , Tubular Heat Exchanger , UHT

DESCRIPTION

The FT94LT comprises a three stage heat exchanger complete with the services, controls and instrumentation required to operate the system. The heat exchanger and control panel are mounted on top of a stainless steel table with feed pump and all services and electrical cabinet located below.

The tubular heat exchanger design uses ten concentric 316 stainless steel tubes for product preheating (two tubes), final heating (four tubes) and cooling (four tubes). In all cases the product passes down the inner tube and the heating/cooling medium passes through the annulus in a counter current direction. As with all Armfield systems, it comes with hygienic fittings throughout as standard; it is easy to clean and very flexible in use.

For the main heating, steam is applied to the service side of the heating section using an electro-pneumatic steam control valve. The product temperature is measured at the end of the heat exchanger (or holding tube). This value is used by a Proportional, Integral and Derivative (PID) control algorithm (implemented in the Programmable Logic Controller (PLC)), to control the steam regulating valve, hence ensuring the user defined set point is maintained.

In order to provide a gentle preheat action, steam at sub-atmospheric pressure (and hence low temperature) is applied to the service side of the preheat section. In this way, steam temperatures at or significantly below 100°C can be produced, and low differentials between steam temperature and product temperature are achieved. Stable preheat temperatures of 60°C or less are feasible. Control of the steam pressure/ temperature is achieved by a manual steam control valve, and automatic PID control is an available option.

Cooling water is applied to the cooling section of the product heat exchanger via a rotameter. For applications requiring chilled exit temperature, the system can be configured to use two initial cooling tubes using city water, and two final cooling tubes using a source of chilled water (Armfield can provide a recirculating water chiller if required).

The system is PLC controlled, with a high resolution TFT 8" colour touch screen panel making it extremely user-friendly to configure and monitor processing parameters. The operator is prompted at every stage whenever intervention is required.

All operation functions are controlled from this panel, including configuration and mode of operation (sterilisation, process or Clean In Place).

Different sets of processing parameters can be edited, stored and quickly recalled using the menu capability of the system.

32 distinct storage locations included as standard.

Similarly, the ancillary items, such as the homogeniser and sterile filler, are also controlled from this panel. The system can be quickly and easily interfaced to other free-standing Armfield process items such as a mixing vessel, a chiller or a sterile filling system.

TECHNICAL SPECIFICATIONS

FT94LT Service Unit Technical Specifications

FEED PUMP

Progressive cavity variable speed pump

Flow rate: 20-100 l/h

Discharge pressure: 18 bar (max)

HEATING SYSTEM

External steam supply

Maximum temp: 165°C

Safety cut outs: Relief valve (7 barg)

Max product heating duty (preheat + heating): 17 kW

TUBULAR HEAT EXCHANGER

Number of tubes:

- 10 2 preheat
- 4 heating
- 4 cooling



TUBE DIAMETER

Inside tube diameter: 8.1 mm

Outside diameter: 22 mm

Length (heated): 1.46 m

Product volume: 1.0 l

Material of construction: 316 Stainless steel

Assembled test pressure: 30 bar

Working pressure: 24 bar (max)

FLOWMETER OPTION (FT94LT-40)

Type: Electromagnetic

Flow range: 20-100 l/h

FEATURES & BENEFITS

- Throughputs from 20 – 100 l/h (dependent on conditions)
- Process temperatures up to 150°C
- Sterilisation In Place (SIP) option
- Cost effective laboratory system
- Compact design and easy installation
- Low product hold up (can operate with as little as 15 litre process batches)
- Suitable for producing large batches of material for extensive product testing
- Rapid start up using steam heating
- Designed for easy integration into a process line
- Links directly to sterile filling bench
- Indirect heating, tube in tube UHT heat exchanger system
- Touch screen control for ease of use
- Continuous operation
- Capable of aseptic operation
- Fully instrumented
- Controlled product preheat option
- Automatic product divert with temperature option
- Hygienic connections
- Built-in Clean In Place (CIP)
- Suction feed with in-line level sensor

Options

DATA LOGGING OPTION (FT94LT-DTA-ALITE)

A sophisticated data logging package enables the measured data to be recorded onto a standard Windows PC (not supplied). The software records all temperatures, product and steam pressures, feed pump speed and many other sensor

values.

Data from optional accessories (e.g. the flowmeter) is also recorded. Data may be displayed in tables and graphs (both real time and comparisons across previous runs), and may also be exported to spreadsheet software through a USB connection.

STATIC MIXERS (FT94LT-21)

The FT94LT-21 is a set of static mixers for two heat exchanger tubes. These significantly improve heat transfer with more viscous products and can be used for improving both heating and cooling efficiency. Multiple sets can be used to optimise both heating and cooling performance.

PRODUCT DIVERT OPTION (FT94LT-22)

The FT94LT-22 Product Divert option enables product that has not been processed to a sufficiently high temperature to be diverted to drain thus ensuring product quality and heat treatment. The divert temperature is adjustable. Diversion is achieved using a low hold-up twin diaphragm valve block operated by compressed air and PLC controlled. Diverted product is cooled in a tubular heat exchanger before being led to drain.

FLOWMETER OPTION (FT94LT-40)

The standard unit displays an estimated flow rate calculated from the feed pump speed. This is accurate enough for many applications, but where more accuracy is required, a flowmeter is available for measuring the product flow rate.

PARTICULATE VALVE ACCESSORY (FT94LT-42)

The FT94LT-42 is a manually controlled pinch valve, which provides much better performance than the standard back pressure valve when used with products containing particulates.

STERILISATION OPTION (FT94LT-45)

Sterilisation is achieved by applying steam onto the outside of the cooling tubes instead of cold water. This sterilises the cooling tubes and gives the power to sterilise a downstream homogeniser. The FT94LT-45 option provides the switching valves and pipework necessary to perform this.

CONTROLLABLE PREHEAT OPTION (FT94LT-46)

This option is required when it is necessary to achieve an accurate and specific preheat temperature (e.g. when it is important to homogenise at a particular temperature), or when using the preheat facility by itself for pasteurising at lower temperatures.

It replaces the standard manual preheat control valve with an automatically controlled electro-pneumatic valve. A PID loop is used to control the temperature to the operator's desired set point by actuation of the valve. The option also includes an electronic pressure sensor to measure the steam pressure. This pressure and its equivalent temperature (determined in the PLC) are displayed on the control panel.

FEED VESSEL (FT94LT-51)

The FT94LT-51 Feed Vessel option is a 50 litre stainless steel vessel with lid, fitted with a product outlet valve and a drain (or recirculation) valve. The assembly is mounted on castors for ease of movement.

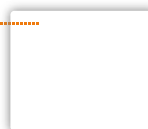
SUCTION FEED WITH AUTO CHANGEOVER (FT94LT-53)

This option allows continuous automatic suction from an external tank with the sensors to detect "no flow" and then switching to water to begin C.I.P.

HOLDING TUBES (FT94LT-60 / 61 / 62)

FT94LT-60 15s at 50l/h

FT94LT-61 30s at 50l/h



Holding tubes can also be provided to suit your holding time and flow rate requirements. Please contact us with your specific requirements.

RECIRCULATING CHILLERS (FT63, FT64)

A recirculating chiller enables product to be output at reduced temperatures. The FT63 is suitable for lower flow rates, but the FT64 is recommended for higher flows

STERILE FILLER (FT83-94)

When used with the sterile options, the FT83 can be used to fill presterilised containers in a sterile environment.

The FT83-94 version is completely compatible with the FT94LT and is controlled from the FT94LT's touch screen. The FT94LT contains all the necessary functionality to sterilise the FT8394. The FT94LT can also be interfaced directly to a sterile sealed bag filling system. Contact Armfield for details.

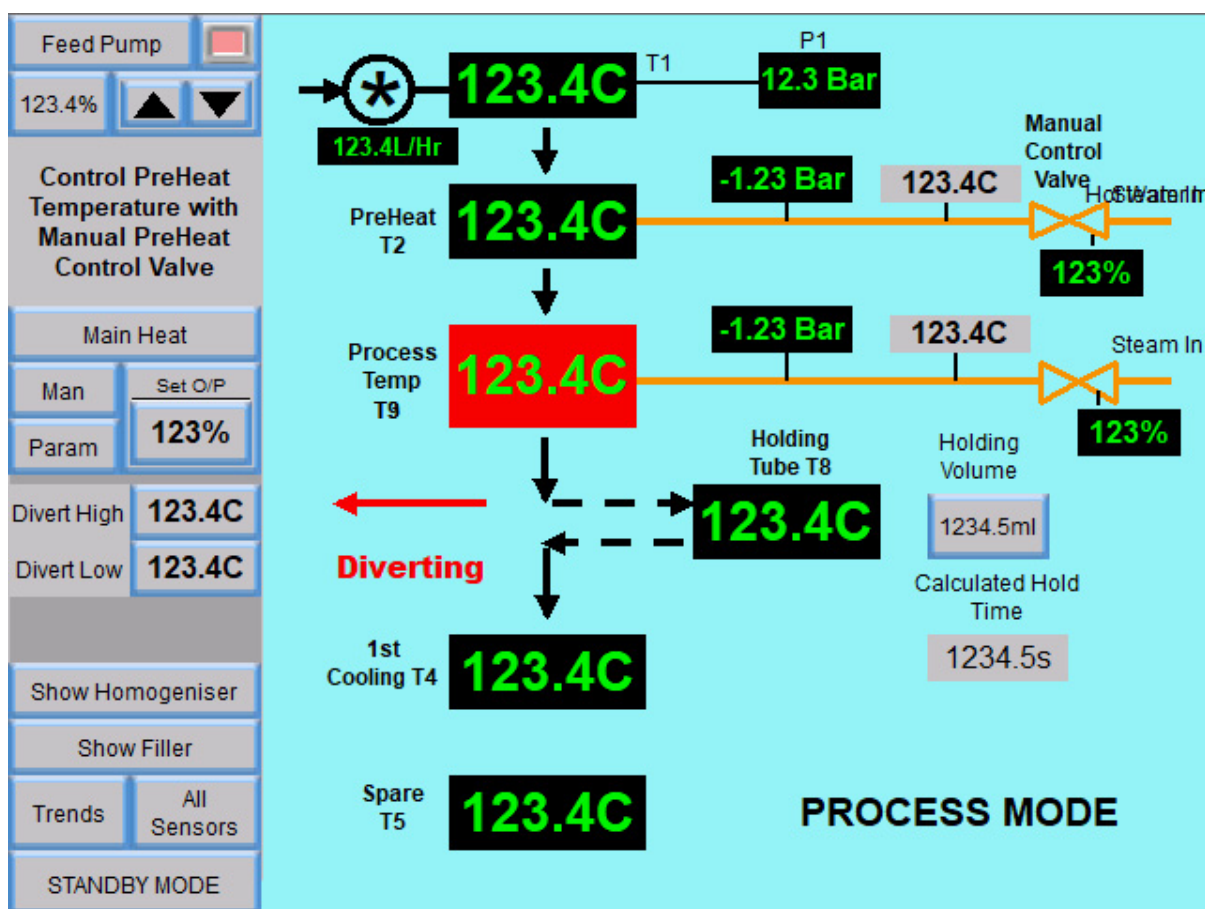
HOMOGENISATION SUB-SYSTEM (FT91-94L)

In-line homogeniser can be provided for use with the FT94LT. Controls for the homogeniser are fully integrated into the FT94LT and its speed can be automatically matched to the FT94LT throughput. The homogeniser is fully sterilisable and can be positioned adjacent to the FT94LT. Both upstream and downstream processing is possible, in addition to stand alone processing.

MIXING TANKS

Armfield can offer an range of mixing vessels with low speed agitators, optional heated jackets and optional high shear mixing. Standard sizes are 50l and 100l. Often used in conjunction with FT94LT-53.

Please contact us with your specific requirements.



Capabilities

- Baby foods
- Beer
- Condiments
- Confectionery
- Culture media
- Desserts and puddings
- Fruit and vegetable purées
- Health and nutritional products
- Fruit juices and cordials
- Gelatine products
- Gravies
- Cream
- Ice cream
- Sauces and soups
- Yoghurts
- Milk
- Pharmaceuticals

Requirements

- **Electricity supply:** Single phase (see ordering codes)
- **Steam supply:** Heat output of 30kW (min 6.0 bar)
- **Compressed air:** 7 l/s (min 7.0 bar, max 10.0 bar)
- **Cooling water:** 10 l/min at 2 bar Chilled water is required for two-stage cooling configurations. **Optional Armfield FT63/FT64 chiller units are available**

Shipping Specifications

Volume: 4.0m³

Gross weight: 425Kg

Dimensions

Length: 1.95m

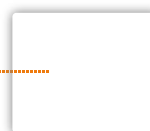
Width: 0.80m

Height: 1.50m

Order Codes

- **FT94LT-A:** 230V/1ph/50Hz (10 A)
- **FT94LT-G:** 230V/1ph/60Hz (10 A)

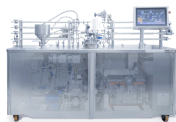
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